

The Theory of the Initial Conditions

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We revisit old ideas with a modern perspective and approach quantum cosmology from a quantum information-theoretic starting point. Our view is that the leap from quantum mechanics in the laboratory frame to the universe as a whole is key to understanding its origins. The problem of the external observer and the conflict of principle between unitarity, causality, and diffeomorphism invariance are explored in bubble nucleation theories. We argue causality is fundamentally incompatible with the assumption that the universe has no external observer. A unique closed-form cosmological wavefunction describing the probability amplitude of the initial state of bubble/antibubble nucleation events is derived. The wavefunction vanishes at the initial singular state, yielding numerous observable consequences. We propose a CMB double-slit experiment to test the theory.

I. MOTIVATION

The cosmos seems to have been born into a spatially flat and thermal state of low entropy, originating from a curvature singularity beyond the effective field theory (EFT) description of space, time, matter, and energy [1]. This grand puzzle demands new physics. The research program presented here aims to discover a unifying principle of the physics of creation and its observable consequences.

The mythos of our shared humanity is a liminal space of creation stories. The origin of the universe is the ultimate question and a timeless puzzle. The philosophy of our program is a form of Neoplatonism that hypothesizes all of reality can be derived from a single unifying principle—an ancient idea. But physicists need more than philosophy and mathematics. We need empirical data, and the empirical data is conclusive: the universe is quantum.

Understanding the nature of the cosmological singularity requires a complete theory of quantum gravity. Indeed, the conceptual problems in the foundations of physics at extremely early times do not begin at the Planck scale [2]. The Copenhagen interpretation of quantum mechanics (QM), which assumes a Heisenberg cut between quantum and classical systems for wavefunction collapse, breaks down when the scale of the universe is on the order of an atom. QM in the ‘laboratory frame’ applied to the early universe as a whole requires the troubling assumption of an external classical observer to prepare the initial state of the cosmos. We argue that Copenhagen can not be the final formulation of QM. A compendium of clues points elsewhere.

One of the most consequential experiments in the foundations of physics is the Aharonov–Bohm effect [3–6]. How can a charged particle’s gauge potential couple to the complex phase of a nonphysical probability wave? Is the measured *change* in phase not an indirect measurement of the physical existence of the wavefunction?

There is still more evidence of the reality of the quantum state. The Pusey–Barrett–Rudolph (PBR) no-go theorem has been experimentally confirmed: either the quantum state is an objective physical state, or the assumption of preparation independence is violated [7–9]. But if the wavefunction is physical, where do probability amplitudes and the Born rule come from? The measurement problem, the problem of the external observer, and the problem of time in quantum cosmology are all deeply related [10]. These foundational conceptual issues can not be ignored in a fundamental theory.

The most cogent formulation of QM with a physical wavefunction is Everett’s seminal work: *The Theory of a Universal wavefunction* [11, 12]. Everett profoundly impacted Wheeler and DeWitt, inspiring the Wheeler–DeWitt (WdW) equation and the field of quantum cosmology [13]. The development of the canonical quantum gravity program of WdW and what became known as the many-worlds interpretation diverged. Work on the foundations of QM became orthogonal to quantum gravity. On the one hand, Everettian QM is done almost exclusively in the non-relativistic limit. Space and time are not just classical. They are Newtonian! On the other hand, the WdW equation is ill-posed.

A fundamental assumption of quantum cosmology is that a Planck-sized universe with closed spatial topology can spontaneously nucleate from nothing, where “nothing” is the absence of matter, radiation, space, and time [14–17]. The Hamiltonian vanishes on physical states with compact geometries in diffeomorphism-invariant theories. In canonical quantum gravity, the wavefunction of the universe satisfies the WdW equation:

$$\hat{H}\psi[h_{ij}, \phi] = 0 \quad (1)$$

where $\hat{H}$ is an infinite-dimensional differential operator in superspace. The wavefunction of the universe $\psi[h_{ij}(x), \phi(x)]$ is defined on the space of all spacelike 3-metrics $h_{ij}(x)$ with field configurations $\phi(x)$. The wavefunction contains all physical information of the system. In principle, $\psi(h, \phi)$ should contain the answer to all physical questions one can ask about the universe. There are an infinite number of possible boundary conditions to

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impose on the equation and, therefore, an infinite number of possible solutions. In standard QM, unique solutions are found by imposing boundary conditions in a physical setup external to the system. There is no laboratory frame external to the universe. To make the WdW equation well-posed, one must specify a fundamental physical law that provides a well-defined boundary condition in superspace.

Most notable are the no-boundary proposal of Hartle and Hawking [18–20] and the tunneling proposal of Vilenkin [21–24]. In the no-boundary proposal, $\psi(h, \phi)$ is given by a Euclidean path integral over compact 4-geometries $g_{\mu\nu}(x, \tau)$ with field configuration $\phi(x)$ bounded by a 3-geometry:

$$\psi(h, \phi) = \int_{\emptyset}^{(h, \phi)} \mathcal{D}g \mathcal{D}\phi e^{-S_E(g, \phi)}, \quad (2)$$

where the "measure" $\mathcal{D}g$ includes integration over the 4-metric, the lapse function $N = N(t)$, and shift vector N^a . The path integral sums over all histories of compact 4-geometries interpolating between "nothing," i.e., a vanishing 3-geometry $\emptyset$ and a finite 3-geometry $h_{ij}(x)$ at some "final time." The boundary principle thus specifies a final 3-geometry h_{ij} and the class of histories to integrate over. One must also determine whether the path integral is performed over 4-manifolds of arbitrary topology or only a restricted class of topologies (and, typically, an inflationary potential as the initial condition). The above integral is violently divergent. The gravitational part of the Euclidean action is unbounded from below. Alternatively, Vilenkin has proposed that $\psi(h, \phi)$ should be obtained by integrating over Lorentzian histories interpolating between a vanishing 3-geometry $\emptyset$ and some final 3-geometry h_{ij} with the so-called tunneling boundary condition, which asserts that ψ only includes outgoing waves at the boundary of superspace. The conditions necessary to specify a unique solution to 1 and the procedure for extracting physically meaningful information from such a solution are far from understood.

There is then the subtle problem of integration over the lapse function N , which determines the total proper time between the initial and final 3-geometries. Should one integrate over only positive values of N , or over all real values of N ? Teitelboim has argued that one should only integrate over $N > 0$ as it introduces a causality constraint that distinguishes the initial and final states, thus preventing one from considering 4-geometries (histories) where the initial and final 3-geometries "conformally invert" into one another [25]. This constraint, however, creates a conflict of principle. In Teitelboim's words: "The causality constraint $N > 0$ disrupts the group structure of four-dimensional diffeomorphisms. Therefore, it appears that in both gravity and supergravity, one is faced with the alternative of preserving either gauge invariance or causality." A series of rigorous calculations by Job Feldbrugge, Jean-Luc Lehners, and Neil Turok shows that from the symmetry $N \rightarrow -N$ in the semiclassical action, there will always be a saddle point corresponding

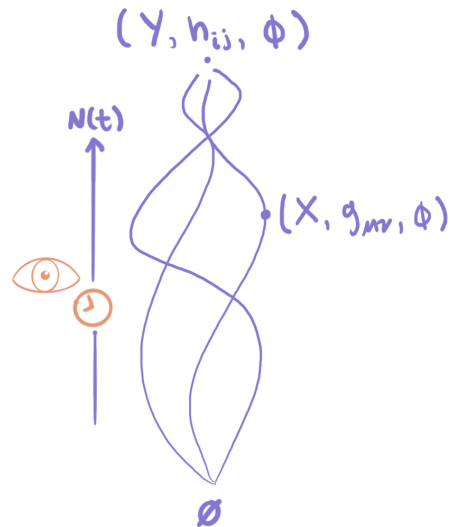


FIG. 1. Illustration of the ill-posedness of the path integral over compact 4-geometries $g_{\mu\nu}$ with topology X bounded by the spacelike 3-geometry h_{ij} with topology Y . $N(t)$ is the lapse function. Spacelike foliations implicitly assume an external observer with a clock measuring the proper time between the initial and final geometries.

to a contracting universe [26–29]. Integrating over all real values of N , one inevitably picks up the saddle points of an expanding and contracting universe, thus obtaining a superposition of amplitudes for the same state evolving forwards and backward in "time" as represented by the scale factor. Such evolution is not unitary and is a striking example of the problem of time. The trio's results strengthen the (now overwhelming) evidence that semiclassical gravity is fundamentally inconsistent.

II. COSMOLOGY FROM QUANTUM INFORMATION

To confront the foundational conceptual and technical challenges raised in this paper, we seek to resurrect some of the ideas of Carl Friedrich von Weizsäcker [30–33] in a bottom-up approach to quantum cosmology. Weizsäcker's *pellicens cogitatio* came to him at a spa in Bad Wildungen in 1954: the quantum theory of binary alternatives, or bits, *is* the theory of objects in 3-dimensional space. Mathematically, he stumbled upon the fact that $SU(2)$, the symmetry group of a quantum bit or qubit, is locally isomorphic to the 3-dimensional rotation group $SO(3)$ of Euclidean space. This was philosophically pleasing to the young Weizsäcker as it explained the apparent three dimensions of space with pure (quantum) logic. He further wondered if all of physics could be reconstructed along similar lines of reasoning. Physics, at its core, reduces to predicting measurement outcomes. Measurement outcomes can be restated in terms of empirically decidable n-fold alternatives. Any

n -fold alternative can be trivially embedded into a Cartesian product of binary alternatives. Thus, physics reduces to information or, more precisely, *potential* information.

Weizsäcker, together with Erhard Scheibe and Georg Süßmann, began with the well-known and deep relationship between tensors and spinors: there always exists a map between tensors of order n and spinors of order $2n$ [34]. Their research program aimed to reconstruct physics from quantum information by elucidating the subtle connections between spin structures and spacetime structure, which would later be subsumed into the broader program of "spinorism" [35–37].

It is well known that the Pauli 4-vector defines a map from Minkowski space to the vector space of Hermitian matrices. One may identify the points $x^\mu = (x^0, x^1, x^2, x^3)$ in Minkowski space with Hermitian matrices:

$$\mathbf{X} := x^\mu \sigma_\mu^{\alpha\beta} = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix} \quad (3)$$

where $\sigma^\mu = (I, \sigma_1, \sigma_2, \sigma_3)$ with the Lorentz metric norm $\|x^\mu\| = \sqrt{\det(\mathbf{X})}$. From this identification, one can write down a lightlike 4-vector k^μ (i.e., $k^\mu k_\mu = 0$) precisely if there is a complex 2-spinor $(\kappa)_{\alpha=1,2}$ such that

$$k^\mu \sigma_\mu^{\alpha\dot{\beta}} = \kappa^\alpha \bar{\kappa}^{\dot{\beta}} \quad (4)$$

where the dotted index denotes complex conjugate components and k^μ is interpreted as the wave four-vector of a photon traveling towards the observer. There is thus a natural map between lightlike 4-vectors and 2-spinors. Modulo a rescaling $\kappa \mapsto a\kappa$ for $0 \neq a \in \mathbb{C}$, this identifies the Celestial sphere as the Riemann sphere: $S^2 \cong \mathbb{CP}^1$, i.e., the space of lines of the lightcone that pass through a point in Minkowski space. This is the exceptional spin isomorphism: $Spin(3,1) \cong SL(2, \mathbb{C})$ identifies the light rays of the Celestial sphere as points on the Riemann sphere. The Celestial sphere that defines the geometry of the observer's sky and the Bloch sphere that defines the geometry of a qubit are mathematically equivalent to the Riemann sphere. Is this duality a mere coincidence or a clue?

Instead of elaborating on the spin structure of Minkowski space and its geometric relationship with quantum information, the trio focused on Weizsäcker's idea of "multiple quantization," a scheme for constructing wavefunctions in Minkowski space from qubits. They discovered the Weyl, Dirac, and Klein-Gordon equations as algebraic identities [34]. Since the spinor unimodular group $SL(2, \mathbb{C})$ does not satisfy a unitary norm, they were motivated to transition from a local spacetime model to a global one with a unitary norm. One again begins with the theory of qubits, but this time satisfying a unitary norm. The symmetry group then contains $SU(2)$, $U(1)$, and complex conjugation. The Lie group manifold:

$$SU(2) \times U(1) \cong S^3 \times S^1 \quad (5)$$

is then a candidate model of the global structure of spacetime. Global cosmic space is compactified to S^3 , and cosmic time is compactified to S^1 . The Lie group manifold implies no distinction between the past and the future. Weizsäcker found this to be highly implausible. The distinction between past and future and the absence of closed timelike curves in nature played a significant role in his philosophy of physics. His philosophical desiderata led him to the global spacetime model with the manifold structure

$$S^3 \times \mathbb{R}^+, \quad (6)$$

and the fundamental assumption that the cosmological wavefunction is a square-integrable function living in the Hilbert space $L^2(S^3)$. This assumption, however, puts space and time on unequal footing. If one includes linear time and instead considers the $L^2(S^3 \times \mathbb{R}^+)$ space of functions, such a causality constraint is inconsistent with diffeomorphism invariance as it implies the Hamiltonian $\hat{H}$ no longer annihilates the quantum state Ψ . Worse still, the motivation for the $U(1)$ was to find a spinorial representation of spacetime satisfying a unitary norm. The causality constraint with the basic expression $\hat{H}\Psi = 0$ is simultaneously inconsistent with unitarity and diffeomorphism invariance.

A. The unitary structure of spacetime

In this section, we explore the path not taken and unravel the spin structure of Minkowski space, as well as its geometric relationship with quantum information, and use it as a foundational starting point for quantum cosmology. We draw inspiration from Wheeler, who advocated for this approach in his paper, entitled "*Information, Physics, Quantum: The Search for Links*," famously epitomized by the phrase "It from Bit" [38].

Quantum information has a rich geometry: the fiber bundle structure of the qubit is the Hopf fibration. The mapping from the unit 3-sphere (embedded in a two-dimensional Hilbert space $\mathbb{C}^2$) to the 2-sphere (Riemann sphere) is the Hopf fibration. Schematically, one writes the fibration as

$$\begin{array}{c} S^1 \hookrightarrow S^3 \\ \downarrow \\ S^2. \end{array} \quad (7)$$

The S^3 lives on the surface of S^2 by the identification of phase circles S^1 in S^3 with points on the S^2 . Every possible state of the qubit lives on the surface of the S^2 . The Hopf fibration has a symmetry group. The action of the unitary group $U(2)$ on $\mathbb{C}^2$ leaves the S^3 invariant. It thus descends to an action on the S^2 by ordinary rotations, carrying fibers into fibers as it commutes with the $U(1)$ action. It is known that the unitary group $U(2)$ is a conformal compactification of Minkowski

space [39, 40]. Equation 3 established that Minkowski space can be represented with 2×2 Hermitian matrices $H(2)$ while 4 shows that null vectors can be represented with the exceptional spin isomorphism. Following Armin Uhlmann [40], the unitary representation $U(2)$ can be constructed with a Cayley transform. Let $\mathbf{X} = \mathbf{X}^{\alpha\beta}$ and $u : H(2) \rightarrow U(2)$ be the Cayley transform,

$$u(\mathbf{X}) := U = \frac{\mathbf{X} - iI}{\mathbf{X} + iI}. \quad (8)$$

The manifold structure of the conformally compactified Minkowski space

$$U(2) \cong (S^3 \times S^1)/\mathbb{Z}_2 \quad (9)$$

is equivalent to a theory of qubits with a unitary norm and a $\mathbb{Z}_2$ quotient. (See Appendix A for the complete derivation.) Here, the S^3 can no longer be considered global cosmic space with positive curvature, but the total space of the Hopf bundle, whose base space is the Celestial sphere with the observer at its center. The conformal compactification does not preserve the causal structure of Minkowski space, as the unitary norm introduces closed timelike curves.

To build a visual intuition of the geometry of the conformal compactification, one can express $U(2)$ as a quotient space [122] by decomposing $U(2)$ into a 4-manifold X and its codimension-1 boundary $Y = \partial X$ by the quotient:

$$U(2) \cong X/Y \quad (10)$$

where

$$X = (B^3 \times S^1)/\mathbb{Z}_2, \quad Y = (S^2 \times S^1)/\mathbb{Z}_2. \quad (11)$$

The S^2 are the cross sections of the observer's sky that form the boundary of a 3-dimensional ball $B^3 \in \mathbb{R}^3$ with the observer at its center. It follows that the boundary Y is a null hypersurface with a degenerate metric. The geometric data on Y is Carrollian: (γ_{ab}, ℓ^μ) where γ_{ab} is the metric on the S^2 cross sections and $\ell^\mu \propto k^\mu$ is the (tangent) null generator, i.e., the lighttrays that all pass through the origin, where the observer lives. The spacetime geometry of the $U(2)$ quotient space is worked out in Appendix B.

1. Unitarity of dynamical spacetime?

Observations show that the global structure of spacetime is not static. It is expanding and accelerating away from a singularity in the past of all lightcones. The naive approach to quantum cosmology is the canonical quantization of a Friedmann-Lemaître-Robertson-Walker (FLRW) universe [41]. The FLRW line element is

$$ds^2 = -N^2(t)dt^2 + a^2(t)d\Omega_3^2, \quad (12)$$

where N is the lapse function and $a(t)$ is the cosmological scale factor of the spatial S^3 . The FLRW spacetime has topology

$$S^3 \times \mathbb{R}_{>0} \quad (13)$$

with a singularity at $t = 0$. The standard approach is to truncate the infinite-dimensional ADM Hamiltonian $\hat{\mathcal{H}}$ to a two-dimensional differential operator $\hat{H}$ in minisuperspace and insert a homogeneous (scalar) matter field ϕ . The cosmological wavefunction $\psi = \psi(a, \phi)$ then has two degrees of freedom. The diffeomorphism constraints are manifestly satisfied by this ansatz with factor ordering of Laplace-Beltrami form. The WdW equation (with units $2G/3\pi = 1$ and $c = 1$) then reads

$$\frac{1}{2} \left(-\frac{\hbar^2}{a^2} \frac{\partial}{\partial a} \left(a \frac{\partial}{\partial a} \right) - \frac{\hbar^2}{a^3} \frac{\partial^2}{\partial \phi^2} + V(a, \phi) \right) \psi(a, \phi) = 0 \quad (14)$$

with the minisuperspace potential,

$$V(a, \phi) = -a + \frac{\Lambda a^3}{3} + m^2 a^3 \phi^2. \quad (15)$$

While the above approximation is much simpler than the full theory, the foundational problems remain. If we were to approach canonical quantization with the observer internal to the system and foliate spacetime into null hypersurfaces, the 3-metric h_{ij} would be degenerate, and the Hamiltonian machinery of the WdW formalism would no longer apply. Even if we were to overcome this limitation of the formalism, the absence of time t in 14 presents yet another conflict of principle, independent of how the spacetime is foliated. By introducing some "intrinsic time" for unitary evolution, whether it is the scale factor a , the scalar field ϕ , or a coordinate t conjugated to a perfect fluid mass, diffeomorphism invariance is broken [42].

B. Bubble nucleation

One can instead consider the nucleation of true vacuum bubbles in a metastable false vacuum with a fixed Minkowski background [84]. After some simplifying assumptions, the nucleating bubble is a round sphere, and the minisuperspace is described by a single degree of freedom, the bubble radius $a(\tau)$. The worldsheet metric of the bubble wall is a (2+1)-dimensional de Sitter space,

$$ds^2 = d\tau^2 - a^2(\tau)d\Omega^2. \quad (16)$$

Quantization is done by promoting the conjugate momentum to the momentum operator $\hat{p}_a = -i\partial_a$ with the assumption of the tunneling boundary condition $|a\hat{p}_a\psi| \gg |\psi|$. The WdW equation then becomes

$$(-\partial_a^2 + U(a)) \psi(a) = 0, \quad (17)$$

where

$$U(a) = (4\pi\sigma a^2)^2(1 - a^2/a_0^2) \quad (18)$$

and $a_0 = 3\sigma/\epsilon$ with σ the bubble wall tension and ϵ the difference between the energy densities of the false and true vacuum. The WdW equation for the bubble has the form of a one-dimensional Schrodinger equation for a zero-energy "particle" with position a moving in the potential $U(a)$.

A longstanding conceptual problem is the location of the observer. If one assumes an external observer performing measurements in a landscape of nucleating bubbles, as quantization implies, how do realistic observers confined to the bubble wall extract observable information from the cosmological wavefunction? If we instead take the observer to be living on the surface of a single bubble, what are the probability amplitudes of? The canonical quantization scheme places the observer worldline external to the worldsheet in the false Minkowski spacetime. Measurements are then reduced to an external observer who prepares the initial state of the bubble and observes the nucleation process. Unitary evolution is with respect to the external observer's time—a manifestly unphysical scenario.

1. Bubble nucleation with internal observers

Consider extending the $U(2)$ spacetime to a conformal compactification of the FLRW spacetime. The null cobordism $\mathcal{I} = \partial X$ can be regarded as a creation operator

$$X : \emptyset \rightarrow \mathcal{I}, \quad (19)$$

where one considers the 4-manifold $X = (B^3 \times S^1)/\mathbb{Z}_2$ as a bubble nucleated from nothing $\emptyset$ and blown up by its null boundary $\mathcal{I} = (S^2 \times S^1)/\mathbb{Z}_2$ at infinity. An immediate consequence is that the spatial topology is flat and time is closed. Bubble nucleation in this setup is of spacetime as seen from an observer at the center of the Celestial sphere that bounds a 3-dimensional ball B^3 in flat space. One can think of the spacetime manifold X as the space of all lightrays that will eventually reach the observer. The lightlike hypersurfaces Y foliate the spacetime into constant radius cross sections that form the observer's sky. Measurement outcomes then reduce to counting photons on Y . This is qualitatively different from nucleation scenarios within the WdW formalism, where the foliation is spacelike, and the metric is nondegenerate. The latter places the observer external to the system, where the former places the observer internal to it.

We analyze the geometry and topology of this nucleation scenario in the geodesic light cone (GLC) gauge in Appendix C and show that the spacetime is geodesically incomplete with past and future singularities. The $\mathbb{Z}_2$ action is no longer free, and the quotient space

$$X/\mathcal{I} \cong (S^3 \times S^1)/\text{ref } \mathbb{Z}_2 \quad (20)$$

fails to be a smooth manifold. It is instead an orbifold

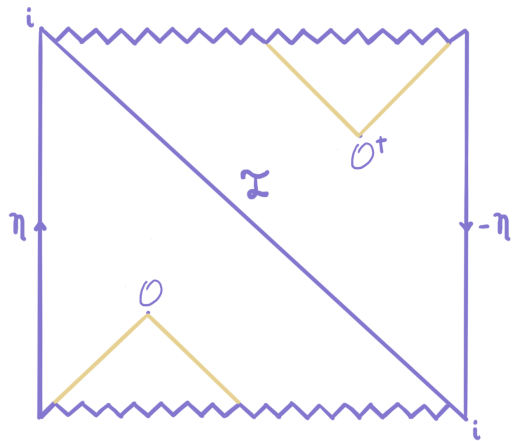


FIG. 2. Penrose diagram of two radiation-dominated FLRW spacetimes with a shared conformal null infinity $\mathcal{I} = (S^2 \times S^1)/\text{ref } \mathbb{Z}_2$, which acts as a lightlike mirror that glues X to its mirror image $X^\dagger$. The time circle identifies past infinity with future infinity, so at the corners where $\mathcal{I}$ becomes singular, timelike and spacelike infinity are identified. The future and past singularities are fixed points (orbifold boundaries) of the reflective $\mathbb{Z}_2$ quotient.

with a reflective $\mathbb{Z}_2$ action. As a consequence, the conformal null infinity $\mathcal{I}$ (in comoving coordinates)

$$ds^2|_{\mathcal{I}} = a^2(\eta)\eta^2 d\Omega^2 \quad (21)$$

acts as a lightlike mirror that glues X ,

$$ds^2|_X = a^2(\eta)(-d\eta^2 + dr^2 + r^2 d\Omega) \quad (22)$$

to its time-reversed mirror image $X^\dagger$,

$$ds^2|_{X^\dagger} = a^2(-\eta)(-d\eta^2 + dr^2 + r^2 d\Omega). \quad (23)$$

The anatomy of $\mathcal{I}$ is a Celestial sphere with the scale factor a labeling its radius: $S_a^2 \cong \mathbb{CP}_a^1$ for $0 < a < \infty$ when $\eta > 0$ and $-\infty < a < 0$ when $\eta < 0$. The corners of the diagram in fig. 2 are manifestly pathological. In the GLC gauge, the Bang is a lightlike mirror that separates two sheets of spacetime by a half-turn π in the null coordinate w and cosmic time η , and we pick up an additional singularity at the observer. The orbifold boundaries are not coordinate-independent.

In a fully realistic cosmology, the $\mathbb{Z}_2$ is a CPT symmetry that reflects matter worldlines $\mathcal{O}$ about $\mathcal{I}$ (or alternatively about $\eta = 0$ in GLC gauge) to antimatter worldlines $\mathcal{O}^\dagger$. The Bang is the nucleation of bubble-antibubble pairs $(X, X^\dagger)$, similar to the proposal of Boyle and Turok [65–67].

C. Compendium of clues

If we take the Borde–Guth–Vilenkin (BGV) theorem [88], and its generalization, the Kinney–Maity–Sriramkumar (KMS) theorem [89] seriously, there is no

past-complete bubble nucleation scenario. In a purely geometric derivation of the BGV theorem, it was shown by KKMS that observers in the same expanding spacetime will not always see the same metric [89]. Consider a stationary observer who sees a net positive amount of expansion with time extending infinitely into both the past and future. Now consider a second observer moving relative to the first. The BGV theorem implies that, as long as the first observer sees a net positive amount of expansion, the moving observer will see a singularity in the finite past. KMS found that the stationary observer sees a flat FLRW metric, while moving observers see an open FLRW metric with a past singularity. From the frame of reference of the stationary observer, the singular metric looks like the nucleation of a spherical bubble expanding outward at the speed of light. Inside the bubble is an infinite, negatively curved universe. Stationary observers see the bubble wall as a spatial boundary, but the moving observer sees the bubble wall as the singularity in their past. The two sets of observers see themselves in different yet overlapping universes!

Time, at least classically, can not be smoothly extended before the Bang. The problem of the initial singularity is deeper still. In “No smooth beginning of spacetime,” Feldbrugge, Lehnert, and Turok have shown that there is no semiclassical formulation of the path integral that is consistent with the sky we observe [28].

Quantizing spacetime implies that lightcones can be superimposed. Numerous theoretical investigations in quantum gravity [43–45], independent thought experiments on *causal indefiniteness* [46–48], and positive experimental data on the reality of indefinite causal order [49–51] all point to causality not being fundamental, but an approximate or emergent feature of the quantum universe. While the evidence accumulates in the direction away from causality, it does not preclude the existence of a *new* fundamental physical principle. Indeed, diffeomorphism invariance is necessary to make sense of any physical theory, but is it sufficient? Quantization schemes—like coordinate systems—are an artifact of formalism. Starting with a classical theory and quantizing it is not what nature does. Nature is quantum, not classical, either from the beginning or without one.

The standard lore in the foundations of physics tells us that degrees of freedom at higher energies (smaller distances) are more fundamental than those at lower energies (larger distances). At the heart of the Wilsonian EFT framework is the assumption that the macroscopic dynamics of a system are independent of the microscopic details. But is this true for gravity? Is the observation of the cosmological constant entering local galaxy dynamics through a universal acceleration scale $a_0 \approx cH_0 \approx c^2 \Lambda^{1/2}$ [52–54] merely a numerical coincidence or a profound clue into the nature of quantum gravity? How is it that something as small as a galaxy “knows” about the de Sitter (dS) radius associated with the whole of the observable universe? UV/IR mixing appears to be a model-independent feature of quantum gravity, with examples

emerging across quantum cosmology, string theory, and beyond [26, 55–72]. Does gravity violate the separation of scales?

Perhaps moving deep into the IR, at the scale of the entire universe, is no less fundamental than working in the UV. This perspective is supported by the data. EFT predicts that the dark energy is dominated by the UV vacuum energy. Yet, it is observed to be astonishingly smaller (by 120 orders of magnitude in Planck units), meaning it sets the IR cutoff that determines the largest scale anyone shall ever see. Motivated by this compendium of clues, we assume reductionism works in both directions. This *universal democracy*, a generalization of Chew’s “nuclear democracy” to cosmology, suggests that the quantum degrees of freedom of the entire universe *are* the fundamental UV degrees of freedom. Intriguingly, this view was discovered at the end of the second superstring revolution with holography [73–77]. The most basic incompatibility between QM and general relativity (GR) is their mutually inconsistent definitions of energy and time. Energy in GR is defined only for geometries that asymptotically approach a well-defined metric tensor at infinity, while in QM, it is defined by the time-translation symmetry of the Hamiltonian. In some deeply mysterious way, the fundamental degrees of freedom exist at the boundary of spacetime—at infinity—where a covariant definition of energy is well-defined in both QM and GR.

One may wonder if the holographic principle [78–80] can rescue Wheeler and DeWitt’s canonical quantum gravity. A recent proposal by Arajua-Regado, Khan, and Wall constructs a holographic theory that lives on Cauchy slices of a Lorentzian bulk in an asymptotically AdS spacetime, where time rather than space is the emergent direction [81]. While they recover bulk unitarity, the theory that lives on the Cauchy slices has negative norm states and is thus non-unitary. Celestial/CFT and dS/CFT dualities, which also have an emergent time dimension, are well known to be non-unitary [82, 83]. Whether one works directly with the WdW equation, sets up a path integral with some boundary data, or constructs a gauge-gravity duality is apparently irrelevant: the conflict of principle between causality, diffeomorphism invariance, and unitarity is independent of the formalism.

The guiding principle of universal democracy motivates the quantum-first program recently proposed by Carroll, which posits that spacetime and quantum fields emerge from the wavefunction of the universe [90, 91]. Carroll and friends [92] approach this program in a non-covariant fashion, leaving a clear gap in the literature. In general covariant theories of quantum gravity, the Hamiltonian annihilates the universal quantum state, and time drops out of the Schrödinger equation. The recent observation of the gravitational Aharonov–Bohm effect [93] further supports the physical reality of the universal quantum state, while the experimental realization of the Page–Wootters mechanism in continuous time [94] is fur-

ther evidence of its timelessness.

Pithily stated: $\hat{H}\Psi = 0$ is the answer to life, the universe, and everything.

D. A quantum theory of bubble nucleation

The leap from quantum mechanics in the laboratory frame to the entire universe is timeless. The logical conclusion is that $\hat{H}\Psi = 0$ has self-contained initial conditions, meaning Ψ is a two-component spinor containing the probability amplitudes of the initial nucleated state of a bubble and antibubble. Probability is strictly frequentist. Every nucleation event that can happen does; some events occur more frequently than others. One can further impose the following criteria on the theory.

- There is no external observer to perform measurements on the state of the bubble: $\hat{H}\Psi = 0$ defines an observer-independent reality and makes no reference to the observer's wordline.
- There are no singularities in the quantum theory: $\Psi = 0$ at the initial singular state.
- All physical information of the universe is contained in Ψ .

1. Codimension-2 holography

The first criterion implies that the pair of angles that parametrize the observer's sky is absent in the theory. In the theory of the initial conditions, the holographic boundary is the codimension-2 surface of the initial data. It is on this surface that Ψ assigns a pair of probability amplitudes to each point. From the last criterion, it follows that the solution to $\hat{H}\Psi = 0$ is unique. Singularity cancellation can happen in only one way: $\Psi = 0$ uniquely at the initial singularity. If Ψ is unique, then so is the potential V in the Hamiltonian $\hat{H}$, which defines the zero-energy state of "nothingness" for spontaneous bubble nucleation to occur.

We construct the surface of the initial data in three steps. First, we begin by Wick rotating $\tau \rightarrow it$ so the conformal null infinity $\mathcal{I}_E = (S_a^2 \times S_\beta^1)_{\text{ref}}/\mathbb{Z}_2$ is Euclidean. The codimension-2 reduction $(S_a^1 \times S_\beta^1)_{\text{ref}}/\mathbb{Z}_2$ is a torus squeezed to a singular point (the initial singularity) that acts as the generator of the reflective $\mathbb{Z}_2$ action. See fig. 3 left diagram. Next, we stage the codimension-2 projective blowup in $\mathbb{C}^2$ with the following Cayley transformations:

$$a \mapsto e^{i\theta} = \frac{a-i}{a+i} \in \mathbb{RP}_a^1 \Rightarrow a = -\cot(\theta/2) \quad (24)$$

and

$$i\beta \mapsto e^{i\phi} = \frac{i\beta-1}{i\beta+1} \in i\mathbb{RP}_\beta^1 \Rightarrow \beta = \cot(\phi/2) \quad (25)$$

where $\beta = 1/T$ is the inverse temperature. The involutions $a \rightarrow -1/a$ and $\beta \rightarrow -1/\beta$ are rotations of π that send $e^{i\theta} \rightarrow -e^{i\theta}$ and $e^{i\phi} \rightarrow -e^{i\phi}$. So the above are one-point compactifications with the thermal projective line $i\beta \in i\mathbb{RP}^1$ and scale projective line $a \in \mathbb{RP}^1$. Finally, we define the surface of the initial data,

$$\mathbb{CP}^1(2,2,2) := \{(e^{i\theta}, e^{i\phi}) \mid 0 \leq \theta < 2\pi, 0 \leq \phi < 2\pi\}$$

with the conformally flat metric

$$ds^2 = d\theta^2 + d\phi^2, \quad (26)$$

where all the curvature concentrates and diverges at the following fixed points from the $\mathbb{Z}_2$ quotient,

$$(\theta, \phi) = (\pi, \pi), (\pi, 0), (0, 0). \quad (27)$$

Modulo the $\mathbb{Z}_2$, the surface of the initial data is a conformal Clifford torus. With the reflective $\mathbb{Z}_2$, the topology is a flat sphere with the above order-2 conical singularities. Specifically, $\mathbb{CP}^1(2,2,2)$ is a flat Riemann sphere with a reflective $\mathbb{Z}_2$ quotient. See the right diagram of fig. 3.

One should think of $\mathbb{CP}^1(2,2,2)$ as an orbifold base space, over which is a generalized fibration of the initial nucleated bubble (or antibubble) $X = (B_a^3 \times S_\beta^1)/\mathbb{Z}_2$ where the thermal circle S_β^1 has radius β and the B_a^3 is the three-dimensional ball of light traveling to the observer in a flat background with scale factor a . The cone points are isolated points over which the topology of X changes. Over $(a, i\beta) = (0, i\infty)$, the thermal circle blows up to infinite radius, and we have $X = [0, i\infty)$. Taking the thermal circle in the $\beta \rightarrow \infty$ limit, and the B_a^3 evaporates away. Over $(a, i\beta) = (0, 0)$ the thermal circle collapses to a singular point and $X = \{0\}$. Hence, $a \propto \beta$. The physical interpretation of the thermal circle is the photon number density n_γ of the B_a^3 .

The initial singularity at $(a, i\beta) = (0, 0)$ is mapped to the conical singularity $(\theta, \phi) = (\pi, \pi)$, while the singularity at the observer $(\theta, \phi) = (\pi, 0)$ is mapped to an observer-independent singularity $(a, i\beta) = (0, i\infty)$, i.e., a state of nothing with zero scale and zero temperature. The remaining singularity in the infinite future $(\theta, \phi) = (0, 0)$ is mapped to $(a, i\beta) = (\infty, i\infty) \equiv \infty$, which is also a state of nothing with infinite scale and zero temperature. Singularity resolution is precisely the condition that $\Psi = 0$ at $(\theta, \phi) = (\pi, \pi)$. If the probability amplitude for an initial singular state is zero, the $\mathbb{Z}_2$ symmetry is free everywhere, and the singularities in the infinite future never occur.

2. Equations of the initial conditions

To derive the equations for the initial conditions, we must endow the surface of the initial data with the appropriate spin structure. Orientable two-dimensional orbifolds admit a spin structure if and only if all cone orders are odd [103]. It follows the orbifold Stiefel-Whitney

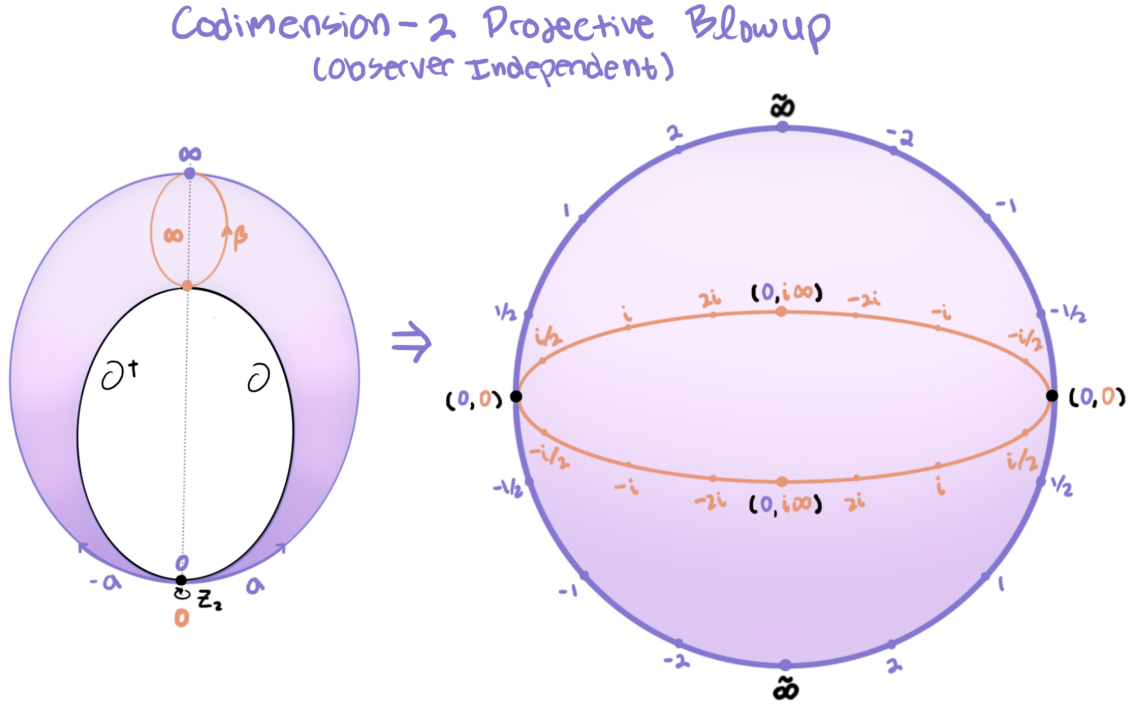


FIG. 3. Left: the Wick-rotated conformal null infinity $\mathcal{I}_E = (S_a^2 \times S_\beta^1) //_{\text{ref}} \mathbb{Z}_2$, pictured with one circle suppressed. The thermal circle S_β^1 undergoes topology change in the limit that $\beta \rightarrow 0$, collapsing to a singular point $\{0\}$ where the only remaining $\mathbb{Z}_2$ action is a reflection. This reflection symmetry is the generator of the conical singularities (the orange and purple dots) in the future. Right: the surface of initial data as the codimension-2 projective blowup to $\mathbb{CP}^1(2, 2, 2) \cong (S_a^1 \times S_\beta^1) //_{\text{ref}} \mathbb{Z}_2 \in \mathbb{C}^2$.

class $w_2^{\text{orb}} \neq 0$, so $\mathbb{CP}^1(2, 2, 2)$ does not admit a standard spin structure. Going around a cone point, spinors pick up a holonomy of -1 , meaning one can globally define spinors only up to sign with the square of the Dirac operator D^2 . The equation of the initial conditions is thus a 2nd-order differential equation of the following form:

$$(D^2 + V(a))\Psi = 0 \quad (28)$$

where Ψ is the projective 2-spinor,

$$\Psi : \mathbb{CP}^1(2, 2, 2) \rightarrow \mathbb{C}^2 \quad (29)$$

that assigns a probability amplitude to the initial nucleated state of a bubble and antibubble, with $V(a)$ an unknown 2×2 matrix potential that defines the vacuum. The surface of the initial data $\mathbb{CP}^1(2, 2, 2)$ is the *boundary of all possible spacetimes* with each bubble/antibubble fundamentally Euclidean. Since $a \propto \beta$ and $\phi = \phi(\beta)$, there is a physical redundancy in the ϕ coordinate. We therefore conjecture that the only consistent Dirac operator is one-dimensional:

$$D = i\sigma_1 \partial_a \quad (30)$$

where eq. 28 becomes a particularly simple 2nd-order ODE.

III. QUANTUM COSMOLOGY FROM THE TOP DOWN

To move beyond mathematical conjecture, we need physical principles. In what follows, we propose the *principles of quantum universality* as the foundation of quantum cosmology and work out their consequences.

- I. There is no external observer who prepared the initial state of the cosmos.
- II. The *form* of the laws of physics does not change with scale.
- III. The universal wavefunction Ψ exists and contains all physical information of the universe.
- IV. Ψ obeys the timeless Schrödinger equation,

$$\hat{H}\Psi = 0 \quad (31)$$

on all scales. Nature has no fundamental scale: Ψ is scale-invariant. Only dimensionless quantities are fundamental.

- V. The universal wavefunction is a 2-spinor Ψ that lives in the Hilbert space of states $L^2[\mathbb{CP}^1(2, 2, 2)] \otimes \mathbb{C}^2$ and assigns a non-zero probability amplitude to every possible initial nucleated state of a bubble and antibubble.

Everything that follows from these principles is pure deduction. The top-down approach is a formal nomology or deductive-nomological model. As it turns out, these principles are both necessary and sufficient to derive and solve eq. 28 for the closed-form solution where $\Psi = 0$ uniquely for an initial singular state.

A. Nomological bootstrap

The simplest possible scale-invariant function is $\Psi \sim a^n$ where $a \in \mathbb{R}$ is the cosmological scale factor and n is an arbitrary integer. One can now solve for the potential V and write down the following one-dimensional "Everett-Wheeler-DeWitt (EWD)" equation:

$$\left(-\frac{d^2}{da^2} + \frac{n(n-1)}{a^2}\right) \Psi = 0. \quad (32)$$

The above equation is not quite consistent with the principles of quantum universality. But observe the following. The potential V is the scalar curvature R of an n -sphere with radius a , the kinetic term is the scalar curvature of an n -hyperbola, and the right-hand side of the equation is the scalar curvature of Euclidean n -space. From this minimal conformal bootstrap, one can derive the fundamental space forms!

GR assumes the equivalence between energy and curvature. Here, it is a consequence. A scalar curvature singularity is at $V(a=0)$. Since the Hamiltonian annihilates the universal quantum state, the scalar curvature singularity cancels with the kinetic term, yielding a universe with zero total energy and global spatial flatness. In GR, the dark energy is proportional to the constant scalar curvature of dS space:

$$R = \frac{d(d-1)}{L^2} = \frac{2d}{d-2}\Lambda \quad (33)$$

where d is the spacetime dimension and L is the radius. The EWD equation has a scalar curvature that scales as $R \sim 1/a^2$. As currently written, Ψ is neither complex nor unitary. The timeless EWD equation occupies a liminal space between geometry and quantum mechanics. Under a change of variables, Eq. (32) is the reduced radial equation of a zero-energy free particle, i.e., the simplest one-dimensional time-independent Schrödinger equation in elementary QM. There is a direct correspondence between $V(a)$ and the effective centrifugal potential $V_{\text{eff}}(r)$ with $n \leftrightarrow -l$, where l is the centrifugal term.

The physical interpretation of the EWD equation is this: the universe, as a whole, can be seen as a Lemaître point-like particle, or Cosmon, with charge M and position a from the origin, sourcing a repulsive potential that accelerates the Cosmon away from the origin and vanishes at infinity. From a geometric perspective, it describes a global charge M that sources the scalar curvatures of an n -sphere and an n -hyperboloid, which together annihilate to flat space. The change of variables

establishes a duality between *position* and *scale*, echoing holography, and that the preferred basis is the scale basis. The equation is timeless, so one can interpret the Hamiltonian as describing the boundary of some compact geometry at future infinity. The number of spatial dimensions $n+1$, as written, is also unfixed.

Can we bootstrap our way to a unitary instantiation of the EWD equation and explain *why* we observe only three spatial dimensions? Yes, we can. There is a trivial action $a \rightarrow -a$ and a non-trivial action $n \rightarrow -n$. The former is a symmetry of the system; the latter is not. Suppose we demand the latter to be a symmetry of the system. The $\mathbb{Z}_2$ action $n \rightarrow -n$ gives the potential $R_{n+1} = \frac{n(n+1)}{a^2}$ and is interpreted as the scalar curvature of an $(n+1)$ -sphere. The potential then becomes,

$$V(a) = \begin{pmatrix} R_{n+1} & 0 \\ 0 & R_n \end{pmatrix} = \frac{1}{a^2} \begin{pmatrix} n(n+1) & 0 \\ 0 & n(n-1) \end{pmatrix} \quad (34)$$

implying Ψ is promoted to a pair of complex numbers, i.e., a spinor. A timeless 2-spinor solution implies an underlying $SU(n=2)$, forcing the self-consistent equations of motion to take the form

$$(D^2 + \hat{V})\Psi = 0 \quad (35)$$

where $D = i\sigma_1\partial_a$ is the one-dimensional Dirac operator and $\hat{V}$ is the promotion of V to a quantum mechanical operator. Before quantization, the components of V are the scalar curvatures of the 3-sphere and 2-sphere. The Hopf fibration uniquely relates these spheres: one can always think of the 3-sphere as an S^1 -fiber over the 2-sphere. $\mathbb{R}^3$ is special insofar as it is a toric fibration of the S^1 -fibers. Since the S^1 -fibers fill all of $\mathbb{R}^3$, scaling every fiber by the scale factor a is equivalent to scaling the Celestial sphere by the same amount. The map $V \rightarrow \hat{V}$ is the quantization of the Hopf bundle in the scale basis.

1. Solution

Assemble the EWD Hamiltonian and its $\mathbb{Z}_2$ dual into the Hamiltonian operator:

$$\hat{H}_n = \begin{pmatrix} -\partial_a^2 + R_{n+1} & 0 \\ 0 & -\partial_a^2 + R_n \end{pmatrix} \quad (36)$$

and rewrite it as

$$\hat{H}_n = -\partial_a^2 I + \frac{1}{a^2}(n^2 I + n\sigma_3). \quad (37)$$

Now, take the most general first-order square root

$$\hat{Q} := \partial_a + \frac{1}{a}(pI + q\sigma_3) \quad (38)$$

with p, q constants. The adjoint operator is

$$\hat{Q}^\dagger = -\partial_a + \frac{1}{a}(pI + q\sigma_3). \quad (39)$$

Taking the square, one obtains,

$$\hat{Q}\hat{Q}^\dagger = -\partial_a^2 I + \frac{1}{a^2}[(p^2 + q^2 + p)I + q(2p+1)\sigma_3]. \quad (40)$$

To reconstruct $\hat{H}_n$ we solve

$$p^2 + q^2 + p = n^2, \quad q(2p+1) = n \quad (41)$$

and find four solutions:

$$(p, q) = (0, n), \quad (-1, -n), \quad (n - \frac{1}{2}, \frac{1}{2}), \quad (-n - \frac{1}{2}, -\frac{1}{2}).$$

The second-order Hamiltonian operator admits four first-order square-root factorizations. Solutions to the first-order equation

$$\hat{Q}\Psi = 0, \quad \Psi = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \quad (42)$$

are automatically solutions to the second-order equation, with general solutions given by

$$\psi_0 \propto a^{-(p+q)}, \quad \psi_1 \propto a^{-(p-q)}. \quad (43)$$

So the four factorizations give four exponent pairs:

$$(0, n) \implies (a^{-n}, a^n) \quad (44)$$

$$(-1, -n) \implies (a^{n+1}, a^{1-n}) \quad (45)$$

$$(n - \frac{1}{2}, \frac{1}{2}) \implies (a^{-n}, a^{1-n}) \quad (46)$$

$$(-n - \frac{1}{2}, -\frac{1}{2}) \implies (a^{n+1}, a^n). \quad (47)$$

Only the first family of traceless solutions respects the $a \mapsto 1/a$ inversion symmetry of the surface of the initial data $\mathbb{CP}^1(2, 2, 2)$. Since the $\mathbb{Z}_2$ symmetry in the scale coordinate is $a \mapsto -1/a$, the unique solutions to eq. 42 are

$$\psi_0 \propto a^{-n} + (-a)^{-n}, \quad \psi_1 \propto a^n + (-a)^n. \quad (48)$$

When n is odd, we have the trivial solution $\psi_0 = \psi_1 = 0$. Observe that for integer-valued n , the above is real-valued. Promoting n from an integer to a real number $n \in [0, 2)$ and identifying the phase angle $\phi = n\pi$, define

$$a^{-\phi/\pi} = e^{-(\phi/\pi)\ln(a)} \quad (49)$$

with the identity,

$$(-a)^{-\phi/\pi} = e^{-i\phi} a^{-\phi/\pi} = e^{-(\phi/\pi)\ln(-a)} \quad (50)$$

and the logarithm defined on the principal branch,

$$\ln(-a) = \ln(a) + i\pi, \quad a > 0. \quad (51)$$

To stay consistent, whenever we take $a \rightarrow -a$, we must also take $i \rightarrow -i$ with the logarithm defined on the conjugate branch. (This is precisely the -1 holonomy around the cone points in figure 6.) Putting everything together, we obtain the unique solution:

$$\Psi(a, \phi) = \begin{pmatrix} (1 + e^{-i\phi})a^{-\phi/\pi} \\ (1 + e^{i\phi})a^{\phi/\pi} \end{pmatrix} \quad (52)$$

with the curvature operator,

$$\hat{V}(a, \phi) = \frac{\phi}{\pi a^2} \begin{pmatrix} \frac{\phi}{\pi} + 1 & 0 \\ 0 & \frac{\phi}{\pi} - 1 \end{pmatrix} \quad (53)$$

of the quantized Hopf bundle. At $\phi = 0$ the curvature operator $\hat{V} = 0$ and the bulk fiber $X = [0, i\infty)$, i.e., a zero temperature $T = 0$ state of nothing. Inspecting the symmetry of the solution, we see

$$\Psi(-1/a, -\phi) = \Psi(a, \phi) \quad (54)$$

which is precisely the periodic symmetry of the surface of the initial data $\mathbb{CP}^1(2, 2, 2)$. The involution $a \mapsto -1/a$ is a free $\mathbb{Z}_2$ in the scale coordinate. In the redundant coordinate, $\phi \rightarrow -\phi$ is not free: it is precisely the reflective $\mathbb{Z}_2$ that exchanges the bubble with the antibubble. One can immediately check that Ψ vanishes at the initial singularity,

$$\Psi(a, \pi) = \begin{pmatrix} (1 + e^{-i\pi})\frac{1}{a} \\ (1 + e^{i\pi})a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (55)$$

as desired. The probability amplitude for an initial singular state is zero. *It is of profound mathematical beauty that $\Psi = 0$ uniquely at the initial singularity $\phi = \pi$ by Euler's identity.*

B. Hilbert space of nucleated states

A key question is the nature of internal observers and how they extract physically meaningful information from Ψ by making measurements on their sky. The manner in which observers ask questions of Ψ does not follow the standard procedure in the laboratory frame. In this subsection, we build the intuition for extracting physical observables.

Unsuppressing the spinor indices $\alpha, \beta = 0, 1$, the universal quantum state ψ_α transforms under the group action of the maximal torus $T_\alpha^\beta = U(1)^2$ as

$$T_\alpha^\beta \psi_\beta = \psi_\alpha^* \quad (56)$$

where

$$T_\alpha^\beta = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix}. \quad (57)$$

Integrating the squared modulus with respect to ϕ over a full turn, we have,

$$\rho_\alpha(a) = \int_0^{2\pi} \psi_\alpha T_\alpha^\beta \psi_\beta d\phi \quad (58)$$

which is interpreted as the *bubble density* as a function of scale. The Born rule is the measure over an infinity of possible initial nucleated states. Indeed, the integral of eq. 58 from $a = 0$ to $a = \infty$ diverges. The bubble density is symmetric under $a \mapsto -1/a$, meaning the components

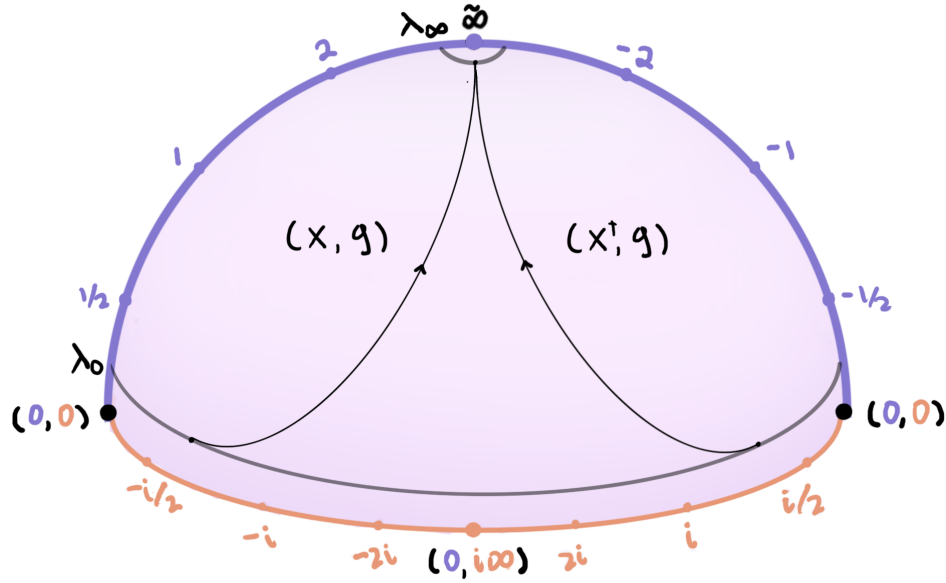


FIG. 4. Surface of the initial data $\mathbb{CP}^1(2,2,2)$ for $-\pi \leq \phi \leq \pi$. At $(a, i\beta) = (0, i\infty)$ we have $\phi = 0$. Internal observers are only defined in the fundamental domain $\lambda_0 < L < \lambda_\infty$; hence, it is only within this domain that the fibration of X and $X^\dagger$ over $\mathbb{CP}^1(2,2,2)$ is physical. The $\phi \rightarrow -\phi$ action takes the bubble X and sends it to its mirror image $X^\dagger$. Time-dependent backgrounds (X, g) and $(X^\dagger, g)$ are curves in the total space that all end on λ_∞ where internal observers decay away. The metric g is thus an observer-dependent field.

of ψ_α need not be integrated over the same domain to integrate over all scales. The domain split places fundamental limits on the information any internal observer can obtain from the system. Normalization of Ψ forces one to define the dimensionless conformal horizon

$$\mathcal{R} = \frac{\lambda_\infty}{\lambda_0}, \quad \lambda_0 < L < \lambda_\infty \quad (59)$$

that places a fundamental limit on the length scales that are observationally accessible in principle. Stated another way, internal observers are emergent and can only exist on length scales L . Defining the scale factor to be in units of length and requiring eq. 35 to be dimensionless, send $a \rightarrow \frac{a}{\lambda_\infty}$ and $a \rightarrow \frac{a}{\lambda_0}$ upstairs and downstairs, respectively. A remarkable duality then follows:

$$\psi_0(\lambda_0, \phi) \psi_0^*(\lambda_0, \phi) = \psi_1(\lambda_\infty, \phi) \psi_1^*(\lambda_\infty, \phi). \quad (60)$$

The UV horizon at λ_0 and the IR horizon at λ_∞ are dual. The Hilbert space of nucleated states is thus defined by the inner product space:

$$\begin{aligned} \langle \psi_0 | \psi_0 \rangle &= \int_{\lambda_0}^{\infty} \int_0^{2\pi} d\phi da \, \psi_0 \psi_0^* \\ &= 4\pi \int_{\lambda_0}^{\infty} da \left(1 - \frac{1}{a^4} \right) \left(\frac{1}{\ln a} \cdot \frac{\pi^2 + 8 \ln^2 a}{\pi^2 + 4 \ln^2 a} \right) \\ &< \infty, \end{aligned} \quad (61)$$

$$\begin{aligned} \langle \psi_1 | \psi_1 \rangle &= \int_0^{\lambda_\infty} \int_0^{2\pi} d\phi da \, \psi_1 \psi_1^* \\ &= 4\pi \int_0^{\lambda_\infty} da (a^4 - 1) \left(\frac{1}{\ln a} \cdot \frac{\pi^2 + 8 \ln^2 a}{\pi^2 + 4 \ln^2 a} \right) \\ &< \infty \end{aligned} \quad (62)$$

and normalized such that $\langle \psi_0 | \psi_0 \rangle + \langle \psi_1 | \psi_1 \rangle = 1$, defining the square-integrable space of functions $L^2[\mathbb{CP}^1(2,2,2)] \otimes \mathbb{C}^2$. We think of λ_0 and λ_∞ as fundamental UV and IR horizons, respectively, beyond which nothing is observable. It is clear that the UV horizon is the Planck length, $\ell_P \equiv \lambda_0$. The IR horizon λ_∞ that determines the largest observable scale is an unknown. The calculation of this number is the subject of the proceeding paper.

IV. OBSERVABLE CONSEQUENCES

One interprets the horizons λ_0 and λ_∞ as the Compton wavelengths of the masses m_0 and m_∞ . Units are restored to eq. 35 by writing down the mass matrix $M = \text{diag}(m_\infty, m_0)$ and multiplying by M^{-1} on both sides. (The absolute values of the masses are not fundamental, as they depend on the choice of unit system. As long as the dimensionless conformal horizon $\mathcal{R}$ is conserved, the physics remains unchanged.) The potential curvature $\hat{V}(a, \phi)$ sources bubble nucleation. In the limit

that $\phi \rightarrow \pi$, the potential curvature (in units of energy)

$$\hat{V}(a) = \frac{\hbar^2}{m_\infty a^2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (63)$$

is sourced entirely by the charge m_∞ . Thus, the heavy charge m_0 is not fundamental but a composite of m_∞ . One thinks of these charges as the fundamental source of spacetime and a realization of Mach's principle, where the fundamental charge density $\rho_\infty = m_\infty/\lambda_0^3$ of empty space is a conserved quantity.

A. Creation

The ψ_0 and ψ_1 components are the probability amplitudes of the initial nucleated bubble X and antibubble $X^\dagger$, respectively. See fig. 4 for the diagrammatic representation of bubble nucleation. Immediately, one sees that the squared modulus has the exchange symmetry: $a \mapsto -1/a \leftrightarrow \phi \mapsto -\phi$ and therefore that $|\psi_0|^2 \gg |\psi_1|^2$ for $a \sim \lambda_0$. For $a \sim \lambda_\infty$, it is the other way around. The initial conditions are asymmetric. This explains the observed matter-antimatter asymmetry.

The potential curvature $\hat{V}(a, \phi)$ contains the geometric data for bubble nucleation with internal observers in the Euclidean 4-manifold X and its antimatter mirror image $X^\dagger$. The topology is lightlike: $X = (B_a^3 \times S_\beta^1)/\mathbb{Z}_2$ with the observer at the center of the Celestial sphere surrounded by thermal radiation. This is the flatness and horizon problem: the initial nucleated bubble is flat, homogeneous, isotropic, and dominated by thermal radiation. But if the initial nucleated bubble is homogeneous and isotropic, then what sourced the inhomogeneities in the early universe that seeded structure formation?

The ultra-light vacuum charge m_∞ sources the repulsive vacuum energy $\hat{V}_{00}$, energetically favoring an inflationary-like scenario of rapid expansion, followed by the slow decay of $\hat{V}_{00}$ as $a \rightarrow \infty$. Since $m_\infty \approx 0$, for sufficiently large scale factors $\hat{V}_{00}$ behaves like a constant. To normalize ψ_α , one must first compute λ_∞ from first principles, and then we can finally calculate the vacuum expectation value,

$$\langle \hat{V}_{00} \rangle \approx \text{Observed Vacuum Energy}. \quad (64)$$

Multiplying this quantity by $m_\infty/\hbar^2$, we have a quantity in units of $1/\text{Length}^2$. This number is expected to be close to the observed cosmological constant Λ . In the $\phi \rightarrow \pi$ limit, the state $\psi_\alpha = 0$, meaning $\hat{V}_{11}$ never vanishes. The charge that sources the $\hat{V}_{11}$ component is the Planck mass m_0 , i.e., the mass of the smallest possible black hole. Thus $\langle \hat{V}_{11} \rangle$ is the expectation value of the energy contribution from spontaneously nucleated primordial black holes.

But if m_0 is a composite of m_∞ , the nucleated black hole mass function must violate decoupling and scale like a^3 to conserve the charge density ρ_∞ since its mass density dilutes like a^{-3} as the bubble expands! We think of

these holes as topological defects of the vacuum orbifold $\mathbb{CP}^1(2, 2, 2)$. Once matter crosses the event horizon, the cosmological coupling decreases to $k < 3$. In general, these vacuum defects will have a cosmological coupling $0 < k < 3$.

Indeed, the source of the thermal radiation is the Hawking temperature,

$$T = \frac{\hbar c^3}{8\pi G M k_B}. \quad (65)$$

The $\hat{V}_{00}$ component is repulsive everywhere while the $\hat{V}_{11}$ component has a decaying attractive phase for $0 < \phi < \pi$, is zero at $\phi = \pi$, and then becomes repulsive for $\pi < \phi < 2\pi$. Most of the probability amplitude concentrates near $(a, i\beta) = (0, i\infty)$ for $\phi > \pi$, meaning the overwhelming majority of initial nucleated states are dominated by ultra-massive cold black holes. A state of almost nothing. Nucleation events near $\phi = \pi$ are dominated by ultra-light hot black holes. Thus, two competing mechanisms set up structure formation after the initial nucleation event and during the quantum roll-down of $\hat{V}$: 1) the nucleation of primordial black holes that either violently evaporate in regions where their Hawking radiation dominates the rate of expansion, or 2) become stable against their Hawking radiation where the rate of expansion dominates.

B. Cosmology as a quantum laboratory

The null boundary $\mathcal{I}$ is a codimension-1 null hypersurface that evaporates away in the remote future when all light signals dilute below what can, in principle, be measured. It is essentially an inside-out event horizon of a black hole. One can apply the area entropy law directly to the Celestial sphere along $\mathcal{I}$ and see that the entropy $S \sim \pi a^2$ is counting all of the states on the Riemann sphere. The microstates are all possible configurations of the observer's sky. This explains the arrow of time: bubbles that nucleate near the Planck scale are in a low-entropy state and become more entropic as they expand.

The theory of the initial conditions is a codimension-2 holographic theory on the orbifold boundary $\mathbb{CP}^1(2, 2, 2)$. We think of $\mathbb{CP}^1(2, 2, 2)$ as the background-independent past boundary of bubble nucleation. Time-dependent backgrounds (X, g) are curves that can begin at any point in the fundamental domain on $\mathbb{CP}^1(2, 2, 2)$, as pictured in fig. 4. From the perspective of internal observers foliating their sky into lightlike hypersurfaces, X is a dynamical fiber. Time-dependent backgrounds are solutions to the bulk field equations.

A fundamental assumption in the derivation of ψ_α was that it contains all physical information of the universe. It follows that the bulk field equations must be equivalent to the theory of the initial conditions. An educated guess on the *form* of the bulk field equations is as follows.

$$\hat{H}_a{}^b \psi_b = (E_g + E_{-g}) \psi_a = 0 \quad (66)$$

where $E_g = -E_{-g}$ is the total gravitational energy, E_{-g} is the total non-gravitational energy, and $a, b = 1, 2, 3, 4$ are spinor indices. The total energy is zero in both theories. Dirac 4-spinors are *reducible* representations that can always be decomposed into a pair of *irreducible* Weyl 2-spinors

$$\psi_a = \begin{pmatrix} \chi_\alpha \\ \zeta_\alpha \end{pmatrix}. \quad (67)$$

Here, they map to the lightlike 4-vectors that foliate the observer's sky in the bubble and antibubble, respectively. Second quantization, i.e., the promotion of ψ_a to an operator-valued field $\hat{\psi}_a$, is a quantum field theory on the lightlike hypersurfaces $(Y, Y^\dagger)$.

1. A cosmic double-slit experiment

Every initial nucleated bubble $X = (B_a^3 \times S_\beta^1)/\mathbb{Z}_2$ is in thermal equilibrium with inverse temperature β and the initial density matrix

$$\hat{\rho}_{ab}(0) = \frac{e^{-\beta \hat{H}_{ab}}}{Z}, \quad Z = \text{Tr}(e^{-\beta \hat{H}_{ab}}). \quad (68)$$

If there is no unique beginning to the universe, then not all observers will agree on the initial conditions. The graviton degrees of freedom, which are described by traceless symmetric matrices, contribute to the off-diagonal components of the density matrix and drive coherences of many overlapping spacetimes. It is not possible, even in principle, to determine which spacetime (X, g) the observer is in, as it would require knowing the precise position and momentum of every particle in the spacetime. Moreover, the phase of ψ_α is *relative*! What is observable is a relative phase shift and a loss of mutual coherence between observers and their skies.

WMAP and *Planck* orbited in the Earth-Sun L2 point in the ecliptic with the Sun orbiting Sagittarius A*. Consider sending a pair of CMB probes to the L2 point—Adam and Eve—and accelerate Adam perpendicular to the ecliptic to reach the galactic rest frame, where he maps the CMB in a stationary state relative to Sagittarius A*. After tracing out the unobserved graviton degrees of freedom, Adam's reduced density matrix is

$$\rho_{\text{CMB}}^{(\text{Adam})} = \text{Tr}_g(U_A \hat{\rho}_{ab}(0) U_A^\dagger), \quad (69)$$

where U_A is the unitary time-evolution operator along Adam's worldline. The angle ϑ between Adam and Eve, relative to the center of Sagittarius A*, is a double-slit in the sky. If, after the acceleration history and all boosts, rotations, translations, and transfer effects are accounted for, and

$$\rho_{\text{CMB}}^{(\text{Adam})} \neq \rho_{\text{CMB}}^{(\text{Eve})}, \quad (70)$$

it would mean that the CMB field does not originate from the same surface of last scattering for both observers.

The key phenomenological observables are the cross-covariance matrix of the CMB multipole coefficients,

$$\Sigma_{\ell m, \ell' m'} = \langle a_{\ell m}^{(\text{Adam})} a_{\ell' m'}^{(\text{Eve})*} \rangle \quad (71)$$

and the angular power cross-spectrum

$$\tilde{C}_\ell = \frac{1}{2\ell + 1} \sum_m a_{\ell m}^{(\text{Adam})} a_{\ell m}^{(\text{Eve})*}, \quad (72)$$

whose magnitude measures coherence and argument measures any net per- ℓ phase offset. Standard Λ CDM predicts the cross-covariance is statistically isotropic and nearly diagonal. If, instead, an interference pattern emerges with off-diagonal components, this would be a smoking-gun signal that quantum mechanics is universal and that there exists a multiplicity of overlapping universes!

V. CONCLUSION

We conclude this paper by motivating the *Morningstar* mission: a precision deep-space worldline observatory to measure the sky and local environment in the galactic rest frame: $\vec{v} \approx -250$ km/s relative to the Sun. *Morningstar* would remove most of the anisotropy in the CMB background, the temperature dipole $T_1 = 3.4$ mK attributed to the motion of the solar system, with a speed of $v \approx 370$ km/s relative to the CMB. Recent measurements of the dipole anisotropy from large-sky radio and quasar surveys have detected an anomalous dipole count that is in a 5.4σ disagreement with the CMB-inferred kinematic dipole [119, 120]. There are then the large-angular-scale anomalies in the CMB temperature, individually ranging from $2 - 4\sigma$, on which both WMAP and *Planck* are in excellent agreement [121]. Since many of these anomalies are statistically independent, they collectively are highly significant and pose a serious challenge to Λ CDM.

Morningstar's core payload would be a cryogenic multi-frequency microwave telescope optimized for large-angle temperature anisotropy, *E*-mode polarization, and searches for primordial *B*-modes, together with an ultra-stable optical atomic clock to measure the proper time along *Morningstar*'s worldline. Its principal observables would be the low- ℓ multipoles and their covariance structure, to be compared directly with the sky measured by a reference observatory near the Earth-Sun L2 point. The secondary payload would be a spectrophotometer to measure the monopole and departures from a perfect blackbody, together with an absolute calibration and metrology platform for ultra-precise systematics tracking, and a foreground astrometry platform optimized for dust characterization. With this instrumentation package, the central hypothesis of a cosmic interference effect is testable at the level of the reduced density matrix of the observed radiation field.

Indeed, the primary challenge is not instrumentation, but payload delivery. Reaching the galactic rest frame presents extraordinary engineering challenges. *Morningstar* would have to be assembled in low Earth orbit and withstand a solar Oberth maneuver with a solar sail assist and a nuclear pulsed drive to reach the galactic rest frame. The scale, complexity, and cost of the mission would exceed that of the International Space Station (ISS), meaning it would have to be a multinational endeavor with broad support. Humanity's first interstellar laboratory would be a profound milestone in our journey to learn of our origins. What better way to bring the nations of the world together and ease geopolitical tensions than our collective aspiration to know our place in the cosmos? What is more, if *Morningstar's* core mission is successful, it would mean there are no paradoxes associated with (interdimensional) time travel.

A. Time travel

We can take our reasoning further still. Fact I: UV/IR mixing is a general feature of the theory. Fact II: an initial nucleated bubble at laboratory scales has a nonzero probability amplitude, although expected to be very small. One can hypothesize the existence of an accessible energy scale for controlled bubble nucleation—at the laboratory scale, with a four-point matter-antimatter reaction—where the interior of the bubble is an extraliminal space with $\hat{V}(a, \phi)$ felt locally, allowing one to go "around" time into the past of some other spacetime (X', g') in the neighborhood of (X, g) by a phase rotation of $e^{i\phi}$. (The phase of eq. 52 is relative and therefore observable!) Such technology could not only be the key to post-scarcity and interstellar travel, but also our final rebellion against nature to survive into the infinite future.

But if time travel is possible, then where are all the time travelers? Could this be the highly advanced "technologies of unknown origin" or unidentified anomalous phenomena (UAP) we are observing in our sky? Are we about to make contact with our descendants from the future?

1. The world that is to come

"From all things, One, and from One, all things," Heraclitus uttered. Unitarity is an ancient idea. From the East, with examples such as *Brahman* ("All is One"), *Ein Sof* ("Boundless One"), *Wahdat al-Wujud* ("Unity of Being"), *Yi Qi* ("One Energy"), to the "The One" of the Gnostics and the pre-Socratic philosophers of the West, humanity has pondered the unity of the flux. When instantiated as a fundamental organizing principle, unitarity gives us the most predictive theory ever discovered and has birthed technological wonders that have transformed our world—for better and for worse. Through-

out all of human history, we have been a deeply divided species. So much so that we have used our knowledge of quantum physics to develop nuclear arsenals capable of annihilating the human species.

But if the deepest truth of quantum mechanics is that we are all One, connected in profound and mysterious ways we have yet to understand, then there is hope for a future where science and spirituality are not separate and serve as the foundation of a new kind of civilization. One that may be co-located with us here on the sacred Pale Blue Dot we call Earth. A civilization wise and knowledgeable enough to understand that without the energy of each and every one of us, from birth to death, equation 66 would not make sense, and the universe could not work. From simple physical principles, forgiveness and love become unconditional. In Lak'ech Ala K'in.

Appendix A: $U(2)$ compactification of Minkowski space

Let $x^\mu = (x^0, x^1, x^2, x^3)$ be the 4-position vector in Minkowski space. Take the inner product with the Pauli vector matrix $\sigma^\mu = (I, \sigma_1, \sigma_2, \sigma_3)$ where I is the 2×2 identity matrix. The resulting matrix is

$$\mathbf{X} := x^\mu \sigma_\mu^{\alpha\beta} = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix} \quad (\text{A1})$$

with Lorentz metric norm $\|x^\mu\| = \sqrt{\det(\mathbf{X})}$. The matrix $\mathbf{X}$ is an element of the space of 2×2 Hermitian matrices $H(2)$. Let $u : H(2) \rightarrow U(2)$ be the Cayley transform,

$$u(\mathbf{X}) := U = \frac{\mathbf{X} - iI}{\mathbf{X} + iI}. \quad (\text{A2})$$

Explicitly, we have,

$$U = \frac{1}{q(x^\mu)} \begin{pmatrix} 1 + \det(\mathbf{X}) + 2ix^3 & 2(ix^1 + x^2) \\ 2(ix^1 - x^2) & 1 + \det(\mathbf{X}) - 2ix^3 \end{pmatrix} \quad (\text{A3})$$

where

$$q(x^\mu) = \det(\mathbf{X}) - 1 + 2ix^0 \quad (\text{A4})$$

and

$$\det(\mathbf{X}) = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \quad (\text{A5})$$

The manifold structure of $U(2)$ is calculated as follows. Let

$$U(2) = \{U \in H(2) \mid UU^\dagger = 1\} \quad (\text{A6})$$

and

$$SU(2) = \{U \in U(2) \mid \det(U) = 1\}. \quad (\text{A7})$$

Clearly, $SU(2) \subset U(2)$. We also have $U(1) \subset U(2)$ as scalar multiplication of the identity I by a complex phase,

$$U(1) = \{e^{i\theta} I \mid \theta \in [0, 2\pi)\}. \quad (\text{A8})$$

Both $SU(2)$ and $U(1)$ are subsets of $U(2)$, but their Cartesian product is not diffeomorphic to $U(2)$ since the intersection is non-empty:

$$SU(2) \cap U(1) = \{U \in SU(2) \mid \det(U) = 1, U = e^{i\theta} I\},$$

where $\det(e^{i\theta} I) = e^{i2\theta} = 1 \implies \theta = 0$ or $\theta = \pi$. So the intersection is

$$\{e^{i \cdot 0} I, e^{i\pi} I\} = \{I, -I\} \cong \mathbb{Z}_2 \quad (\text{A9})$$

which must be quotiented out to avoid overcounting. Since $U(1) \cong S^1$ and $SU(2) \cong S^3$, we have

$$U(2) \cong (S^3 \times S^1)/\mathbb{Z}_2 \quad (\text{A10})$$

which completes the proof.

Appendix B: $U(2)$ spacetime geometry

We work in geodesic lightcone coordinates: $x^\mu = (\tau, w, \theta, \phi)$ where $\tau = \text{const.}$ are spacelike hypersurfaces, $w = \text{const.}$ are lightlike hypersurfaces, i.e., the past (or future) lightcones of the observer, and $\theta^a \equiv (\theta, \phi)$ are the regular spherical angles on the observer's sky at any moment. The metric on X is then

$$ds^2|_X = dw^2 - 2dw d\tau + \gamma_{ab} d\theta^a d\theta^b \quad (\text{B1})$$

where

$$\begin{aligned} \gamma_{ab} d\theta^a d\theta^b &= (w - \tau)^2 d\Omega^2 \\ &= (w - \tau)^2 (d\theta^2 + \sin^2 \theta d\phi^2) \end{aligned} \quad (\text{B2})$$

and time is compactified into a circle with the helical and antipodal identifications, respectively:

$$(\tau, w) \sim (\tau + T, w + L), \quad x^\mu \sim -x^\mu. \quad (\text{B3})$$

For simplicity, choose the *temporal gauge* where the observer's world line $\mathcal{O}$ is a circle located at

$$w_o(\tau) = \tau_o \pmod{L}, \quad (\text{B4})$$

where τ_o is the reception time of a photon emitted from elsewhere towards the observer. In general, we use the "o" subscript to denote evaluation at the observer world line and the "0" subscript to denote evaluation at any constant w . The intersection of the $\tau = \text{const.}$ and $w = \text{const.}$ hypersurface is a spatial 2-dimensional surface, i.e., the S^2 cross sections whose radius is the travel time τ_o of a photon to reach $\mathcal{O}$. The spatial distance to the observer's world-line is then

$$r := \pm(w - \tau_o), \quad (\text{B5})$$

so evaluation at the observer world-line gives $r_o = 0$, where the θ^a coordinates are not defined by definition. The null boundary Y foliates the spacetime into $w =$

$w_0 \pmod{L}$ hypersurfaces. The metric on Y is degenerate:

$$ds^2|_Y = (w_0 - \tau)^2 d\Omega^2, \quad (\text{B6})$$

i.e., there is no $d\tau$ term. To recover the full $U(2)$ spacetime fix a periodic conformal defining function $\Omega(w) = \sin(2\pi w/L)$ with a simple zero at $w \equiv 0 \pmod{L}$. The conformally rescaled metric $ds^2 \rightarrow \Omega^2(w) ds^2$ extends smoothly to the conformal null infinity

$$\mathcal{I} = \{w \equiv 0 \pmod{L}, (S^2 \times S^1)/\mathbb{Z}_2\}, \quad (\text{B7})$$

on which the induced metric vanishes, thereby collapsing $\mathcal{I}$ to a unique point where future infinity, past infinity, and spacelike infinity become the same: $i_\pm = i_0 = i$, completing the quotient space action of $U(2)$. All null geodesics pass through the conformal infinity i . See fig. 5 for a conformal diagram of the quotient space.

Appendix C: Conformal compactification of the flat FLRW spacetime

Suppose the observer is not rotating and thus defined as an irrotational congruence of timelike geodesics, i.e., the family of observers with synchronized clocks to measure the "cosmic time" τ . The general expression of the line element in GLC coordinates [85–87] then reads

$$ds^2 = \Upsilon^2 dw^2 - 2\Upsilon dw d\tau + \gamma_{ab} d\theta^a d\theta^b \quad (\text{C1})$$

where $\Upsilon = \Upsilon(\tau, w, \theta^a)$ is the GLC lapse function. Any spacetime with non-rotating observers, at least locally, can be brought to this form.

Let us now move to the *cosmic time gauge* with η_o the conformal travel time of a photon to reach $\mathcal{O}$. If we assume the observer's sky is homogeneous and isotropic, the solution is given by

$$\tau = t, \quad \Upsilon = a(\eta), \quad w = r + \eta \quad (\text{C2})$$

with

$$\begin{aligned} \gamma_{ab} d\theta^a d\theta^b &= a^2(\eta) r^2 d\Omega^2 \\ &= a^2(\eta) r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \end{aligned} \quad (\text{C3})$$

where $a(\eta)$ is the cosmological scale factor and $d\eta = dt/a$. Plugging the above into C1, the metric on X in comoving GLC coordinates is

$$ds^2|_X = a^2(\eta) (dw^2 - 2dw d\eta + r^2 d\Omega^2). \quad (\text{C4})$$

The metric on $Y : w = w_0$ is degenerate:

$$ds^2|_Y = a^2(\eta) r_0^2 d\Omega^2, \quad (\text{C5})$$

where the observer is at the center of a 3-dimensional ball $B^3 \in \mathbb{R}^3$ with radius r_0 filled with lightrays ℓ^μ where $k^\mu \propto \ell^\mu$ is the wave four-vector of a photon traveling towards the observer.

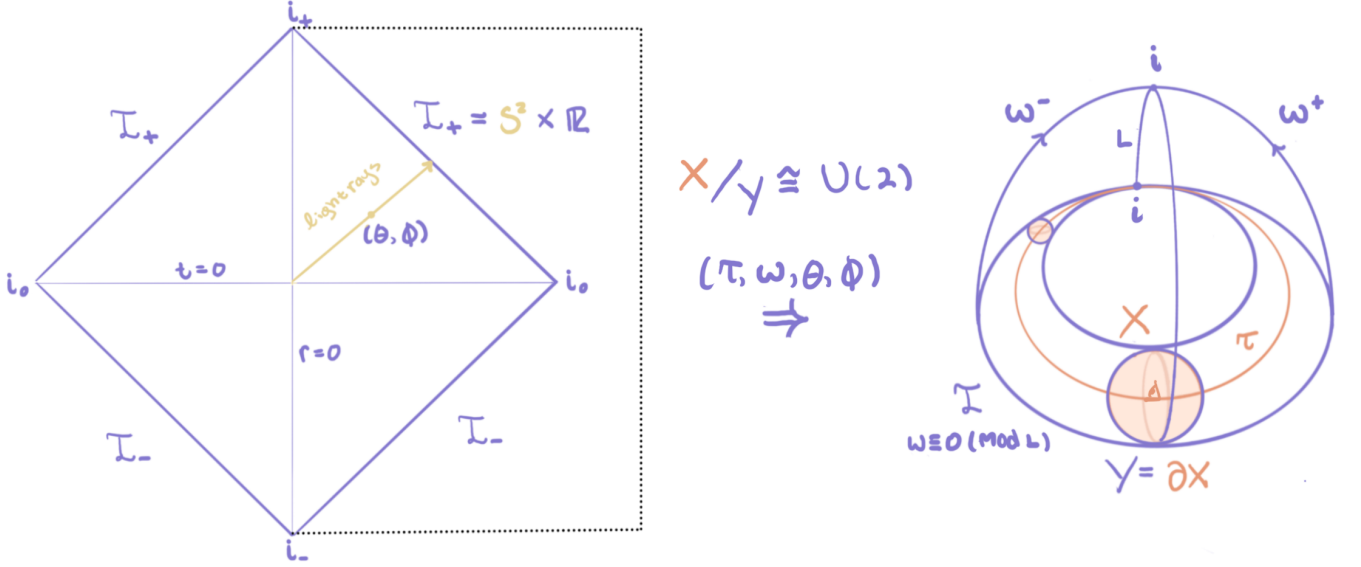


FIG. 5. Diagrammatic representation of the conformal compactification of Minkowski space to $U(2)$. Left: Penrose diagram with future and past infinity identified. Right: $U(2) \cong X/Y$ quotient diagram in geodesic lightcone coordinates with temporal gauge. At the base is the conformal null infinity $\mathcal{I} = (S^2 \times S^1)/\mathbb{Z}_2$, where the S^2 is the Celestial sphere with the observer at its center. Past and future lightcones are labeled by w^- and w^+ , respectively. Going up the null coordinate w at some $\tau_o = \text{const.}$ are all the lightrays emitted from elsewhere with the reception time τ_o to reach the observer. Collapsing $\mathcal{I}$ to the point $i = i_{\pm} = i_o$ is the quotient space action.

It is straightforward to see why the time circle is unphysical for a flat spatial topology. Take the null normal to $Y : w = w_0$,

$$n_\mu = \partial_\mu w = (0, 1, 0, 0). \quad (\text{C6})$$

The (τ, w) block of the bulk metric and its inverse is

$$g_{\mu\nu}|_{\tau,w} = \begin{pmatrix} 0 & -a \\ -a & a^2 \end{pmatrix}, \quad g^{\mu\nu}|_{\tau,w} = \begin{pmatrix} -1 & -1/a \\ -1/a & 0 \end{pmatrix}. \quad (\text{C7})$$

Raising the index of the null normal gives the null generator:

$$\ell^\mu = g^{\mu\nu} n_\nu = \pm(1/a, 0, 0, 0) = \pm(1/a)\partial_\tau, \quad (\text{C8})$$

where the sign ambiguity denotes past or future-directedness of the lightcone. The time circle implies that all timelike curves are closed, where ℓ^μ is everywhere future-directed or everywhere past-directed. The flat FLRW spacetime is not time-symmetric: the de Sitter future $\lim_{a \rightarrow \infty} k^\mu = 0$ can not be identified with the past singularity $\lim_{a \rightarrow 0} k^\mu = \infty$ in any physically meaningful way. At $\eta = 0$, the spacetime collapses to an instantaneous spacelike surface, meaning the $\mathbb{Z}_2$ action is no longer free.

Consider the time-reversal of X , i.e., flipping the sign of ℓ^μ . Define the time-reversal as the adjoint null cobordism $X^\dagger : Y \rightarrow \emptyset$. The composition $X^\dagger \circ X : \emptyset \rightarrow \emptyset$ is the double:

$$D(X) := X \cup_Y X^\dagger = (S^3 \times S^1)/\mathbb{Z}_2 = U(2), \quad (\text{C9})$$

which is again the conformally compactified Minkowski space. From the Heegaard splitting property, the S^3 can be obtained by gluing a pair of 3-balls along their shared boundary. Since the 4-manifold X is the Cartesian product of a 3-ball and a circle, $U(2)$ has a representation of the null boundary Y gluing X to $X^\dagger$ (fig. 6, left diagram). To visualize the gluing procedure, imagine placing a mirror at the touch-point (the conformal infinity i) between each cap. Recall the $\mathbb{Z}_2$ action acts antipodally everywhere. Now identify the points on the boundary 2-spheres to obtain the 3-sphere. The $\mathbb{Z}_2$ action is free (smooth) at the touch-point. To see this, note that the conformal null infinity $\mathcal{I}$ is identified at both ends. Rotate the top cap by a half turn and slide it down until there is only one $\mathcal{I}$. The endpoints are antipodally matched, hence the $\mathbb{Z}_2$ action is free everywhere. The resulting "egg" diagram is the double $D(X)$ with the mirror placed at $\mathcal{I}$. The quotient space X/Y and the double $D(X)$ are equivalent and complementary ways of representing and visualizing the $U(2)$ spacetime.

The Carrollian data on the null boundary Y for the flat FLRW spacetime, i.e., where $\Upsilon = a(\eta)$, is the same as the $U(2)$ spacetime, modulo the physical rescaling of Y . In the $U(2)$ compactification, time is compactified by the helical and antipodal identifications, $(\tau, w) \sim (\tau + T, w + L)$ and $x^\mu \sim -x^\mu$, respectively. The conformal null infinity $\mathcal{I}$ is collapsed to the conformal infinity i by an unphysical periodic conformal rescaling of the metric. The observed expansion of the universe cannot be gauged away, meaning the scale factor $a(\eta)$ is a physical rescaling of the metric. This forces the conformal compactification

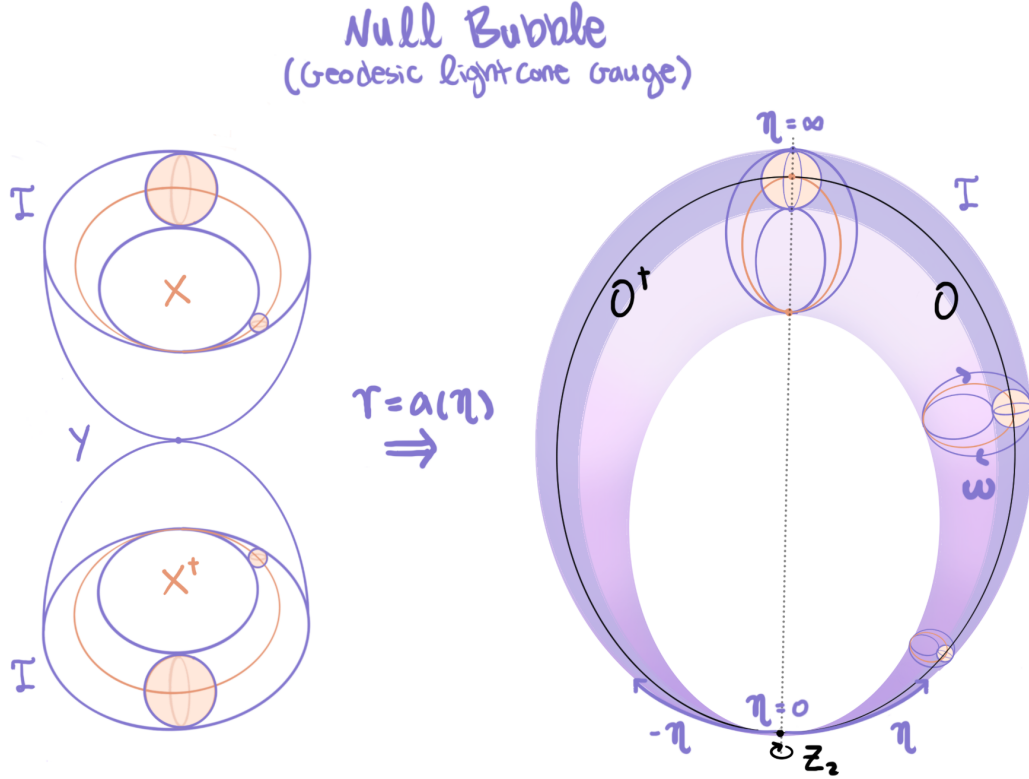


FIG. 6. Left: diagrammatic representation of $U(2) = (S^3 \times S^1)/\mathbb{Z}_2$ as the union of the 4-manifold X and its time-reversed adjoint $X^\dagger$ joined along their shared null boundary $\mathcal{I}$. The $\mathbb{Z}_2$ action is the free antipodal map. Right: diagrammatic representation of the one-point compactification ($\eta \in \mathbb{R}P^1$, $w \in \mathbb{R}P^1$) of the flat FLRW spacetime from the perspective of the observer at the center of a nucleated bubble. At $\eta = 0$ the only remaining $\mathbb{Z}_2$ action is the reflection of the null generator ℓ^μ , which produces a pair of fixed points at future infinity. The 2-sphere at null infinity is the Celestial sphere. The observer is at the apex of the lightcone and is lifted to the center of the Celestial sphere by the local antipodal action of $\mathbb{Z}_2$, which forms the observer's worldline $\mathcal{O}$ and the adjoint observer's worldline $\mathcal{O}^\dagger$.

to be the one-point compactification of time and the null coordinate: ($\eta \in \mathbb{R}P^1$, $w \in \mathbb{R}P^1$). The null boundary $\mathcal{I}$ collapses to a singularity at $\eta = 0$, implying the only remaining $\mathbb{Z}_2$ action is the reflection of the null generator ℓ^μ (fig. 6, right diagram), which generates two additional fixed point singularities at future infinity. An immediate consequence is that the quotient space $X/\mathcal{I} \cong (S^3 \times S^1)_{\text{ref}}/\mathbb{Z}_2$ fails to be a smooth manifold and is instead an orbifold [99, 100]. The conformal null infinity is

$$\mathcal{I} = \{w \equiv 0 \pmod{2\pi}, (S^2 \times S^1)_{\text{ref}}/\mathbb{Z}_2\} \quad (\text{C10})$$

with metric

$$ds^2|_{\mathcal{I}} = (a\eta)^2 d\Omega^2. \quad (\text{C11})$$

There is a double-cone singularity at $\eta = 0$ and a pair of conical singularities at $\eta = \infty$.

Appendix D: De Sitter limit and its anti-De Sitter dual

The calculation of the dS limit is similar in spirit to Jacobson's [105] and Verlinde's [106] thermodynamic

derivation of GR. If one assumes that internal observers can only access at most

$$Z_{\text{obs}} \sim e^{2\pi\mathcal{R}} \quad (\text{D1})$$

qubits, where $\mathcal{R} = \lambda_\infty/\lambda_0$ defines the dimensionless radius of the conformal horizon and the maximum entropy is $S = 2\pi\mathcal{R}$, one can find dS space in the following limit. First, set the dark energy as a constant by setting

$$\Lambda = \frac{1}{\lambda_0\lambda_\infty}. \quad (\text{D2})$$

One can now calculate the radius of the cosmic event horizon R_{CEH} by equating the Gibbons-Hawking entropy to the above:

$$\begin{aligned} \frac{\pi R_{\text{CEH}}^2}{\lambda_0^2} &= 2\pi \frac{\lambda_\infty}{\lambda_0} \\ R_{\text{CEH}} &= \sqrt{2\lambda_0\lambda_\infty} \\ &= \sqrt{\frac{2}{\Lambda}} \\ &\approx 14.26 \text{ billion ly.} \end{aligned} \quad (\text{D3})$$

It is now straightforward to calculate the vacuum energy density (in reduced Planck units):

$$\rho_\Lambda := \rho_\infty = \frac{m_\infty c^2}{\lambda_0^3}. \quad (\text{D4})$$

Now consider a flat acceleration surface with $2\pi k_B$ entropy per unit (Planck) area $1/\lambda_0^2$. Now multiply this quantity by the Hawking temperature $T(g)/\lambda_0$ per unit length of the accelerating surface, where g is the acceleration. The result is the pressure

$$p_0 = \left(\frac{c^2}{G\lambda_0} \right) g, \quad (\text{D5})$$

where the quantity in parentheses is an *area density* with units $[\text{mass}][\text{area}]^{-1}$. The physical construction is an accelerating membrane, like the surface of a glowing soap bubble. The above equation is the ideal gas law $pV = k_B T$, where $V = \lambda_0^3$ is the unit volume swept out by the membrane. The above is a thermodynamic derivation of $F = m_0 g$, Newton's second law. Now consider the dual acceleration surface with $4\pi k_B$ entropy per unit R_{CEH}^2 and repeat the above steps to obtain:

$$p_\infty = \left(\frac{\Lambda \hbar}{c\lambda_0} \right) g, \quad (\text{D6})$$

which is just $F = m_\infty g$ when multiplied by λ_0^2 on both sides. The above pressures obey the ideal gas law, meaning we construct the stress-energy tensor $T_{\mu\nu} = \text{diag}(\rho_\infty, p_\infty, p_\infty, p_\infty)$. In the UV limit, one obtains the vacuum stress-energy tensor,

$$T_{\mu\nu}^{(\text{vac})} = \lim_{g \rightarrow -c^2/\lambda_0} T_{\mu\nu} = \begin{pmatrix} \rho_\Lambda & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} \quad (\text{D7})$$

where the sign on g distinguishes the expanding and contracting states.

The first fact we learned from the top-down bootstrap is that energy and curvature are equivalent, a fundamental assumption of GR. There are not many options to

equate $T_{\mu\nu}^{(\text{vac})}$ to curvature on the left-hand side. Moreover, since all physical information is contained in Ψ , the fact that it is a power law explains why the power spectrum is a power law. We also learned that the form of the laws of physics is independent of the chosen coordinate system. No coordinate system was used to solve eq. (35). In this section, we learned that the fundamental charge of the vacuum, the gravitational charge m_∞ , is equivalent to inertial mass. These deductions are none other than the equivalence principle and diffeomorphism invariance! Given these deductions, the simplest equations that follow are none other than Einstein's field equations:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu} \quad (\text{D8})$$

where $R_{\mu\nu}$ is the Ricci curvature tensor, R is the scalar curvature, $g_{\mu\nu}$ is the metric tensor, and κ is the Einstein gravitational constant. There is then the dual vacuum stress-energy tensor for the contracting state. As before, we can construct a stress-energy tensor $T_{\mu\nu} = \text{diag}(\rho_0, p_0, p_0, p_0)$ for an ideal gas. In the IR limit,

$$T_{\mu\nu}^{(\text{dual-vac})} = \lim_{g \rightarrow c^2/\lambda_\infty} T_{\mu\nu} = \begin{pmatrix} \rho_0 & 0 & 0 & 0 \\ 0 & -p & 0 & 0 \\ 0 & 0 & -p & 0 \\ 0 & 0 & 0 & -p \end{pmatrix}. \quad (\text{D9})$$

One obtains a bizarre substance with the Planck energy density and a positive vacuum energy pressure $p = \Lambda$ (with the minus sign absorbed) that satisfies the equation of state for dark matter: $w = p/\rho_0 \approx 0$.

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